

Backward heat martingale, square-coupled charge, and Cartan reduction

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claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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Purpose and scope

This note continues the A13 strict-past proof without creating a new peer claim. It corrects the remaining charge by retaining the completed square, removes the apparent shellwise heat-current drift by a terminal-backward Doob martingale, and proves that the finite low block and the high-frequency counterterm secants are not the final obstruction. The complete finite-cutoff problem is reduced to one averaged raw-current/Cartan secant with its positive Gauss–Newton square retained.

Closed here. Exact square-coupled and Gibbs–Pythagoras identities; the terminal-backward heat martingale and controlled telescope; uniform $C^{1,1}$ heat regularity; exact scalar shellwise no-gos; the averaged production frame/Cartan secant; a finite-low-block bound; and a summable high-shell covariance-trace loss using the control energy once.

Still open. A cutoff-uniform lower bound for the remaining averaged raw-current/Cartan–Jacobi secant under every regular declared strict-past control. Consequently no controlled-shell one-use theorem, Nelson moment, interacting measure, regulator removal, or tier promotion is asserted.

All exact identities are finite-dimensional and cutoff-wise. The analytic tail bounds use the hash-pinned $L = 16$ A1 symbol, the common real-even sharp spectral shells, one strict-past control block per mutually orthogonal shell, the fixed positive density floor, and the six-real convention. A general progressive covariance flow which revisits the same Fourier range is not silently identified with this ordered block class. The controlled identities and bounds are stated for bounded cylindrical controls, or more generally for strict-past controls for which every displayed product is integrable. An extension to arbitrary finite-energy drifts by localization and lower semicontinuity belongs to the successor gate and is not asserted here.

1 The full score and the square that must remain

Use the A11–A13 strict-past notation. Conditional on $\mathcal{F}_{j-1}$, the exact shell increment is

$$F_j(u) = I_j(u) + C_j(u) = \frac{1}{2}\{u^T T_j u - \text{Tr } T_j\} + \langle s_j, u \rangle + N_j(u), \quad s_j = \ell_j + m_j, \quad (1.1)$$

where $T_j \geq 0$, $C_j(u) = \langle m_j, u \rangle + N_j(u)$, and $N_j(0) = DN_j(0) = 0$. The A11 telescope says

$$V_J^{\text{ren}} = \sum_{j \leq J} F_j. \quad (1.2)$$

If $\mathcal{K}_j u = (K_j u, DK_j u)$ denotes the physical value/derivative displacement, then more is true:

$$F_j(u) = V_j^{\text{ren}}(Z_{j-1} + \mathcal{K}_j u) - V_{j-1}^{\text{ren}}(Z_{j-1}). \quad (1.3)$$

Consequently

$$\langle s_j, v \rangle = DV_j^{\text{ren}}(Z_{j-1})[\mathcal{K}_j v]. \quad (1.4)$$

Thus the full score is conservative at the energy-gradient level. The non-exact target-space current one-form found earlier is still present inside the frame representation; it is not a curl of the complete shell energy.

Let $q > 0$,

$$R_j = (\mathbf{I} + qT_j)^{-1}, \quad H_j = T_j + q^{-1}\mathbf{I}, \quad (1.5)$$

and let ξ_j be standard Gaussian. Put

$$\overline{F}_j(h) = \mathbb{E}_j F_j(\xi_j + h), \quad \overline{N}_j(h) = \mathbb{E}_j N_j(\xi_j + h), \quad (1.6)$$

$$P_j(h) = \frac{1}{2} \left| H_j^{1/2}(h + qR_j s_j) \right|^2. \quad (1.7)$$

Exact metric completion gives

$$\boxed{\overline{F}_j(h) = -\frac{|h|^2}{2q} - \mathcal{S}_j^\sharp(h), \quad \mathcal{S}_j^\sharp(h) = \frac{q}{2} \langle s_j, R_j s_j \rangle - \overline{N}_j(h) - P_j(h).} \quad (1.8)$$

The square P_j is part of the charge. Bounding $q\langle s_j, R_j s_j \rangle/2 - \overline{N}_j$ alone is a strictly stronger statement and repeats the earlier source-square error. At $h = 0$,

$$P_j(0) = \frac{q}{2} \langle s_j, R_j s_j \rangle, \quad \mathcal{S}_j^\sharp(0) = -\mathbb{E}_j N_j(\xi_j) = -\mathbb{E}_j F_j(\xi_j). \quad (1.9)$$

After averaging and summing, (1.2) gives

$$\mathbb{E} \sum_{j \leq J} \mathcal{S}_j^\sharp(0) = -\mathbb{E} V_J^{\text{ren}}(X_J) = 0. \quad (1.10)$$

Any shellwise positive-part or absolute-value estimate destroys this exact base-law cancellation.

Conditional Gibbs–Pythagoras identity

Assume first that the displayed exponential normalizer is finite and that the shifted Gaussian has finite relative entropy against the conditional tilted law. The identities below then extend to the usual localized/truncated class whenever both sides are finite.

Define

$$\kappa_j = \frac{1}{q} \log \mathbb{E}_j e^{-qF_j}, \quad \nu_j(\mathrm{d}u) = e^{-qF_j(u) - q\kappa_j} \gamma_j(\mathrm{d}u). \quad (1.11)$$

Then

$$\boxed{\mathcal{S}_j^\sharp(h) = \kappa_j - \frac{1}{q} H(\gamma_h \mid \nu_j).} \quad (1.12)$$

Indeed,

$$H(\gamma_h \mid \nu_j) = \frac{|h|^2}{2} + q\overline{F}_j(h) + q\kappa_j, \quad (1.13)$$

and (1.12) follows from (1.8). Completing the conditional Gaussian also gives the exact non-separated normalizer

$$\begin{aligned} \kappa_j = & -\frac{1}{2q} \log \det {}_2(\mathbf{I} + qT_j) + \frac{q}{2} \langle s_j, R_j s_j \rangle \\ & + \frac{1}{q} \log \mathbb{E}_{\mathcal{N}(-qR_j s_j, R_j)} e^{-qN_j}. \end{aligned} \quad (1.14)$$

For a triangular shifted law Q and the sequentially normalized law $P = \prod_j \nu_j$, the entropy chain rule yields

$$\mathbb{E}_Q V_J^{\text{ren}} = -\frac{1}{2q} \mathbb{E}_Q \sum_j |\hat{h}_j|^2 - \mathbb{E}_Q \sum_j \kappa_j + \frac{1}{q} H(Q | P). \quad (1.15)$$

Here $\hat{h}_j$ is the conditional mean as a function of the *state* past. If $Z_j = \xi_j + h_j(\xi_{<j})$, then

$$\hat{h}_j(z_{<j}) = h_j(\mathcal{T}_{<j}^{-1} z_{<j}). \quad (1.16)$$

The executable negative control shows that replacing (1.16) by $h_j(z_{<j})$ changes the chain identity. Also, P is not generally the global Gibbs law: its past-dependent conditional normalizers do not telescope. Thus (1.15) is an exact geometric identity, not a Nelson estimate.

Fast quartic carrier: factor four is reassembled

For the scalar principal carrier $f(A) = cA^4$, $c \geq 0$, one has

$$T = 2cA^2, \quad \ell = m = 2cA^3, \quad s = 4cA^3, \quad (1.17)$$

$$N(u) = cu^2(u^2 + 4Au + 5A^2) = cu^2\{(u + 2A)^2 + A^2\} \geq 0. \quad (1.18)$$

With $R = (1 + qT)^{-1}$ and $P(u) = (T + q^{-1})(u + qRs)^2/2$, direct expansion gives

$$\frac{q}{2}s^2R - N(u) - P(u) = f(A) - f(A + u) - \frac{u^2}{2q}. \quad (1.19)$$

The ratio $s^2/\ell^2 = 4$ is therefore exactly neutralized by the nonlinear remainder and retained square. At $u = -A$ the right side is $cA^4 - A^2/(2q)$, so a shellwise terminal-only estimate is still false for large A . Equation (1.19) is pointwise. It must not be confused with the Gaussian-averaged charge in (1.8); for example $\mathcal{S}^\sharp(0) = -c(5A^2 + 3)$ in the standard scalar Gaussian fixture.

2 Terminal-backward heat martingale

Let $(g_j, d_{j,i})$ be the value and i th derivative fields of the j th Gaussian shell. At each spatial point assume

$$\mathbb{E}[g_j \otimes g_j] = \Sigma_j, \quad \mathbb{E}[d_{j,i} \otimes d_{j,i}] = \Delta \Gamma_{j,i}, \quad \mathbb{E}[g_j \otimes d_{j,i}] = 0. \quad (2.1)$$

Joint Gaussianity and the pinned common-even covariance make the value and derivative variables independent at the same point. Write

$$\Gamma_{j,i} = \sum_{k \leq j} \Delta \Gamma_{k,i}, \quad \Sigma_{r:s} = \sum_{k=r}^s \Sigma_k. \quad (2.2)$$

For a fixed terminal shell J , define the backward coefficient and energy

$$B_j^{(J)} = P_{\Sigma_{j+1:J}} B, \quad (2.3)$$

$$W_j^{(J)}(x, y) = \frac{1}{2} \sum_i \int B_j^{(J)}(x) : (y_i^{\otimes 2} - \Gamma_{j,i}) \, dr. \quad (2.4)$$

The integral in (2.4) is the spatial integral; the heat semigroup acts in the six-real target variable at each point.

Theorem 2.1 (backward heat martingale). For every finite cutoff and every deterministic past state (x, y) ,

$$\mathbb{E}_j W_j^{(J)}(x + g_j, y + d_j) = W_{j-1}^{(J)}(x, y). \quad (2.5)$$

If (a_j, b_j) is $\mathcal{F}_{j-1}$ -measurable, then

$$\mathbb{E}_j W_j^{(J)}(x + a_j + g_j, y + b_j + d_j) = W_{j-1}^{(J)}(x + a_j, y + b_j). \quad (2.6)$$

Proof. Independence in (2.1) factors the same-point expectation. The derivative average changes $y_i^{\otimes 2} - \Gamma_{j,i}$ into $y_i^{\otimes 2} - \Gamma_{j-1,i}$, while the value average changes $B_j^{(J)}$ into $P_{\Sigma_j} B_j^{(J)} = B_{j-1}^{(J)}$. This proves (2.5). Conditioning on the strict past makes (a_j, b_j) deterministic, which proves (2.6). $\square$

Let the controlled state satisfy

$$Z_j = Z_{j-1} + a_j + g_j, \quad DZ_j = DZ_{j-1} + b_j + d_j, \quad Z_0 = DZ_0 = 0. \quad (2.7)$$

Since $W_j^{(J)} = V_j^{\text{ren}}$ and $W_0^{(J)}(0, 0) = 0$, the tower property gives the exact controlled discrete Ito identity

$$\boxed{\mathbb{E} V_J^{\text{ren}}(Z_J) = \mathbb{E} \sum_{j \leq J} \left\{ W_{j-1}^{(J)}(Z_{j-1} + a_j, DZ_{j-1} + b_j) - W_{j-1}^{(J)}(Z_{j-1}, DZ_{j-1}) \right\}.} \quad (2.8)$$

This is the nonlocal reorganization missing from the forward conditional heat-current calculation. Every uncontrolled $O(1)$ -per-shell heat drift is removed before an inequality is applied.

3 Why the forward heat term cannot be bounded shellwise

The production coefficient is globally $C^{1,1}$ uniformly in the positive floor. If the floor is $e > 0$ and $X = \sqrt{e} Y$, then

$$B_e(\sqrt{e} Y) = e B_1(Y), \quad D_X^2 B_e(\sqrt{e} Y) = D_Y^2 B_1(Y). \quad (3.1)$$

The rational B_1 is smooth on bounded sets and is asymptotically degree two with a smooth angular coefficient. Hence

$$L_B := \sup_{e>0} \sup_X \|D^2 B_e(X)\| < \infty. \quad (3.2)$$

Taylor's formula gives

$$\|P_{\Sigma} B(X) - B(X)\| \leq \frac{L_B}{2} \text{Tr } \Sigma. \quad (3.3)$$

This correct estimate is not summable after taking an absolute contraction with $DX^{\otimes 2} - \Gamma_{< j}$: sharp dyadic scaling gives $\text{Tr } \Sigma_j \asymp 2^{-j}$ and $\text{Tr } \Gamma_{< j} \asymp 2^j$.

The failure is exact already for the scalar coefficient $B(x) = x^2$. Take

$$\sigma_j = 2^{-j}, \quad \gamma_{< j} = 2^j. \quad (3.4)$$

At $x = Dx = h = 0$ the score and completed square vanish, but

$$\mathbb{E}_j C_j = -\frac{1}{2} \sigma_j \gamma_{< j} = -\frac{1}{2} \quad (3.5)$$

for every shell, whereas

$$\mathbb{E} |g_j|^6 = 15 2^{-3j}. \quad (3.6)$$

For a Gaussian past derivative the centered quadratic in (3.5) has standard deviation $1/\sqrt{2}$, again independent of j . Thus shellwise absolute values, positive parts, or sextic domination are impossible.

The executable production audit confirms that this is not an artifact of the scalar model. It computes the exact A1 sharp-cube value/derivative covariances and averages the full rational six-real B at two Gauss–Hermite orders. The origin deficits are

	$N = 64$	$N = 128$	
GH 7	$2.8680745298 \times 10^{-4}$	$2.8526808367 \times 10^{-4}$	
GH 9	$2.8689027298 \times 10^{-4}$	$2.8535053764 \times 10^{-4}$	(3.7)

The registered assertions are the conservative interval $2.84 \times 10^{-4} < -\mathbb{E}C_{128} < 2.87 \times 10^{-4}$, a relative 64/128 gap below two percent, and an order-seven/order-nine relative gap below 5×10^{-4} at both cutoffs. This is cutoff- and quadrature-stable finite evidence for the pinned production coefficients, not an interval proof of a continuum limit.

There is also an analytic zero-floor diagnostic. Write $R = |\psi_1|^2 + |\psi_2|^2$ and $\rho = R + |\psi_3|^2$. Direct Pauli algebra gives

$$\mathrm{Tr} B_e = 12(a + 2b + c_{\mathrm{II}})R - \frac{8bR^2}{\rho + e} - \frac{4c_{\mathrm{II}}R^2(\rho + 2e)}{(\rho + e)^2}. \quad (3.8)$$

Set $e = 0$ first (equivalently, take the high-variance limit $e/c_{\mathrm{sh}} \downarrow 0$). If $\mathbb{E}\psi\psi^* = c_{\mathrm{sh}}\mathbf{I}_3$ and the one-direction derivative covariance is $d\mathbf{I}_3$, Pauli identities then give

$$\mathbb{E} \mathrm{Tr} B = 6c_{\mathrm{sh}}(4a + 6b + 3c_{\mathrm{II}}), \quad (3.9)$$

so the three-direction origin deficit coefficient is

$$\frac{3}{4}d\mathbb{E} \mathrm{Tr} B \simeq 0.163265625 c_{\mathrm{sh}}d. \quad (3.10)$$

At the production values, $b, c_{\mathrm{II}} > 0$, so (3.8) is no smaller at $e > 0$ than at $e = 0$. Hence (3.10) is also a positive-floor lower-bound coefficient, although equality is reserved for the zero-floor limit. The product $c_{\mathrm{sh}}d$ is scale zero. Equations (3.5)–(3.10) explain why the backward martingale, rather than a sharper modulus estimate in (3.3), is load-bearing. The executable fixed-floor computation (3.7), rather than the zero-floor equality (3.9), is the finite-floor numerical evidence.

4 Averaged frame secant and Cartan form

Recall the production frame

$$B(X) = \sum_{A=1}^3 M_A(X) Q_{\mathrm{II}} M_A(X)^T, \quad M_A = [p_A, v_A], \quad Q_{\mathrm{II}} \succeq 0, \quad (4.1)$$

$$p_A = 2S_A X, \quad v_A = 2(S_A - q_A \mathbf{I})X, \quad q_A = \frac{X^T S_A X}{|X|^2 + e}. \quad (4.2)$$

Heat averaging preserves both positivity and the frame representation:

$$B_{j-1}^{(J)}(x) = \int \sum_A M_A(x+r) Q_{\mathrm{II}} M_A(x+r)^T d\mu_{j:J}(r), \quad (4.3)$$

where $\mu_{j:J} = \mathcal{N}(0, \Sigma_{j:J})$. The variable r in (4.3) is a target-space heat dummy held fixed under spatial differentiation; it is not identified pathwise with the actual future Gaussian field.

For a stage state (x, y_i) and physical control (a, b_i) put

$$w_{A,i,r} = M_A(x+r)^T y_i, \quad (4.4)$$

$$\Delta w_{A,i,r} = M_A(x+a+r)^T(y_i+b_i) - M_A(x+r)^T y_i, \quad \Delta M_{A,r} = M_A(x+a+r) - M_A(x+r). \quad (4.5)$$

Direct expansion of (2.4) gives the exact averaged secant

$$\begin{aligned} \Delta W_{j-1}^{(J)} = \sum_{i,A} \int \int \{ & w_{A,i,r}^T Q_{\Pi} \Delta w_{A,i,r} + \frac{1}{2} \Delta w_{A,i,r}^T Q_{\Pi} \Delta w_{A,i,r} \\ & - \text{Tr}(Q_{\Pi} M_A(x+r)^T \Gamma_{j-1,i} \Delta M_{A,r}) \\ & - \frac{1}{2} \text{Tr}(Q_{\Pi} \Delta M_{A,r}^T \Gamma_{j-1,i} \Delta M_{A,r}) \} dx d\mu_{j:J}(r). \end{aligned} \quad (4.6)$$

The second term is the retained positive Gauss–Newton square. The last term is negative; its scale will be shown summable rather than assigned a false sign.

For the non-exact current one-form

$$\omega_A = dm_A - q_A d\rho, \quad K_A(X) = X^* \omega_A, \quad (4.7)$$

the prior exact identity remains valid under the heat average:

$$K_A(x+a+r) - K_A(x+r) = d_x \int_0^1 \omega_A(x+r+ta)[a] dt + \int_0^1 d\omega_A(x+r+ta)[a, y+tb] dt, \quad (4.8)$$

$$d\omega_A = \frac{d\rho \wedge dm_A}{\rho + e}, \quad \|d\omega_A\|_{\text{op}} \leq 4. \quad (4.9)$$

Thus (4.6) is precisely an averaged exact-current plus Cartan-curvature secant. In second-variation language, if $w_t = M_A(x+r+ta)^T(y_i+tb_i)$, then

$$\frac{d^2}{dt^2} \frac{1}{2} |w_t|_{Q_{\Pi}}^2 = |\dot{w}_t|_{Q_{\Pi}}^2 + \langle w_t, Q_{\Pi} \ddot{w}_t \rangle. \quad (4.10)$$

The first term is positive; the second is the signed Cartan–Jacobi term. They may not be estimated independently at the critical regularity.

5 The covariance-trace channels are summable

Let $\Lambda_j = 2^j$, $a_j = K_j h_j$, and $b_{j,i} = D_i K_j h_j$. The A1 symbol lower bound and the sharp-cube count imply, after enlarging constants for finitely many low shells,

$$\|K_j\|_{\mathcal{H}_j \rightarrow L^2} \leq C_K \Lambda_j^{-2}, \quad \|DK_j\|_{\mathcal{H}_j \rightarrow L^2} \leq C_G \Lambda_j^{-1}, \quad (5.1)$$

$$\max_i \|\Gamma_{j-1,i}\|_{\text{op}} \leq C_{\Gamma} \Lambda_j. \quad (5.2)$$

Here $M_R := \sup_{\Lambda,n} |R_{\Lambda}(n)|$ is the common multiplier bound, and $c_{\text{sym}} > 0$ is a pinned high-frequency symbol constant such that $\lambda_{\min} A(k) \geq c_{\text{sym}} \langle k \rangle^4$ on every shell mode. One admissible set is

$$C_K = 4M_R c_{\text{sym}}^{-1/2} \alpha^{-2}, \quad C_G = 2M_R c_{\text{sym}}^{-1/2} \alpha^{-1}, \quad (5.3)$$

$$C_{\Gamma} = L^{-3} M_R^2 c_{\text{sym}}^{-1} \alpha^{-2} \left(24 + \frac{\pi^2}{3} \right), \quad \alpha = \frac{2\pi}{L}. \quad (5.4)$$

For (5.4), group lattice points by $\|n\|_{\infty} = m$, use $\#\{\|n\|_{\infty} = m\} = 24m^2 + 2$, and bound $k_i^2 \langle k \rangle^{-4} \leq (\alpha m)^{-2}$.

The production frames have floor-uniform deterministic bounds

$$|M_A(z)|_F \leq \sqrt{20}|z|, \quad |M_A(z+a) - M_A(z)|_F \leq \sqrt{68}|a|. \quad (5.5)$$

To see the second bound, write $r_A = (S_A - q_A \mathbf{I})z$. Since $S_A^2 \leq \mathbf{I}$ and $m_A = q_A(|z|^2 + e)$,

$$|r_A|^2 \leq |z|^2, \quad \nabla q_A = \frac{2r_A}{|z|^2 + e}, \quad |z| |Dq_A(z)[a]| \leq 2|a|. \quad (5.6)$$

Hence $|Dp_A[a]| \leq 2|a|$ and $|Dv_A[a]| \leq 8|a|$, proving $\sqrt{2^2 + 8^2} = \sqrt{68}$.

Let S_j denote the positive magnitude of the last trace term in (4.6):

$$S_j = \frac{1}{2} \sum_{i,A} \int \int \text{Tr}(Q_{\Pi} \Delta M_{A,j,r}^T \Gamma_{j-1,i} \Delta M_{A,j,r}) \, dx \, d\mu. \quad (5.7)$$

Equations (5.1), (5.2), and (5.5) give pathwise

$$S_j \leq C_S \Lambda_j^{-3} \|h_j\|^2, \quad C_S = \frac{9}{2} \|Q_{\Pi}\|_{\text{op}}(68) C_{\Gamma} C_K^2. \quad (5.8)$$

Thus

$$\sum_{j \geq j_0} S_j \leq C_S 2^{-3j_0} \sum_{j \geq j_0} \|h_j\|^2. \quad (5.9)$$

For the mixed trace term put

$$L_j = \sum_{i,A} \int \int \text{Tr}(Q_{\Pi} M_A(x+r)^T \Gamma_{j-1,i} \Delta M_{A,j,r}) \, dx \, d\mu, \quad (5.10)$$

and

$$A_j^2 = \int \|Z_{j-1} + r\|_2^2 \, d\mu_{j:J}(r). \quad (5.11)$$

Then

$$|L_j| \leq C_L \Lambda_j^{-1} \|h_j\| A_j, \quad C_L = 9 \|Q_{\Pi}\|_{\text{op}} \sqrt{20} \sqrt{68} C_{\Gamma} C_K. \quad (5.12)$$

It is unnecessary to identify r with a future controlled field. Bounded total value covariance, (5.1), and shell orthogonality give directly

$$\begin{aligned} \mathbb{E} A_j^2 &\leq 2\mathbb{E} \|Z_{j-1}\|_2^2 + 2L^3 \text{Tr} \Sigma_{j:J} \\ &\leq 4\mathbb{E} \|X_{<j}\|_2^2 + 4\mathbb{E} \left\| \sum_{k < j} K_k h_k \right\|_2^2 + C \\ &\leq C_A \left(1 + \mathbb{E} \sum_k \|h_k\|^2 \right). \end{aligned} \quad (5.13)$$

The last line uses the mutually orthogonal one-shot shell ranges pathwise; it is not a future-field substitution. Cauchy–Schwarz over probability and shells therefore yields

$$\mathbb{E} \sum_{j \geq j_0} |L_j| \leq C 2^{-j_0} \left\{ 1 + \mathbb{E} \sum_k \|h_k\|^2 \right\}. \quad (5.14)$$

Combining (5.9)–(5.14), for every $\eta > 0$ one may choose a fixed j_0 large enough that, uniformly in J ,

$$\mathbb{E} \sum_{j \geq j_0} (|L_j| + S_j) \leq \eta \mathbb{E} \sum_k \|h_k\|^2 + C_{\eta, j_0}. \quad (5.15)$$

This spends the control energy once. It does not use the desired Nelson moment.

6 The finite low block and the exact remaining form

Fix the threshold j_0 selected above. Summing (2.8) only over $j < j_0$ gives exactly

$$\mathbb{E} \sum_{j < j_0} \Delta_j^{\text{ctrl}} = \mathbb{E} W_{j_0-1}^{(J)}(Z_{j_0-1}, DZ_{j_0-1}). \quad (6.1)$$

Positivity of $B_{j_0-1}^{(J)}$, quadratic growth of B , bounded total value covariance, and the fixed matrix Γ_{j_0-1} imply

$$W_{j_0-1}^{(J)}(z, Dz) \geq -C_{j_0}(1 + \|z\|_2^2), \quad (6.2)$$

uniformly in J . In the declared orthogonal one-shot shell class, $Z_{j_0-1} = P_{< j_0} Z_J$. The fixed low projection is bounded from L^6 to L^2 , and scalar Young gives, for every $\zeta > 0$,

$$\mathbb{E} \sum_{j < j_0} \Delta_j^{\text{ctrl}} \geq -\zeta \mathbb{E} \|Z_J\|_6^6 - C_{\zeta, j_0}. \quad (6.3)$$

If later controls are allowed to revisit the low range, the equality $Z_{j_0-1} = P_{< j_0} Z_J$ fails and (6.3) needs an extra control term. No such broader progressive-flow theorem is claimed here.

Define the remaining high-shell raw-current secant

$$\mathcal{R}_j = \sum_{i,A} \int \int \{w_{A,i,r}^T Q_{\Pi} \Delta w_{A,i,r} + \frac{1}{2} \Delta w_{A,i,r}^T Q_{\Pi} \Delta w_{A,i,r}\} dx d\mu_{j:J}(r). \quad (6.4)$$

For the regular control class fixed in the scope paragraph, equations (2.8), (4.6), (5.15), and (6.3) prove the strongest closed reduction:

$$\boxed{\mathbb{E} V_J^{\text{ren}}(Z_J) \geq -\zeta \mathbb{E} \|Z_J\|_6^6 - \eta \mathbb{E} \sum_j \|h_j\|^2 - C_{\zeta, \eta, j_0} + \mathbb{E} \sum_{j \geq j_0} \mathcal{R}_j.} \quad (6.5)$$

The constant is uniform in the terminal cutoff J . Thus neither the forward heat drift, the finite low block, nor the high covariance trace is the remaining ultraviolet obstruction.

The exact successor theorem is a lower bound for $\mathbb{E} \sum_{j \geq j_0} \mathcal{R}_j$ with arbitrarily small remaining control and sextic allocations. It must use (4.8)–(4.10) while retaining the positive square in (6.4). The elementary inequality

$$w^T Q \Delta w + \frac{1}{2} |\Delta w|_Q^2 \geq -\frac{1}{2} |w|_Q^2 \quad (6.6)$$

is forbidden: summing its right side repays the full past current energy at every shell.

7 What A9 imports, and what it cannot supply

The backward coefficient $B_{j-1}^{(J)}$ is positive and satisfies

$$\|B_{j-1}^{(J)}(z)\|_F \leq \beta_B(|z|^2 + \kappa_0). \quad (7.1)$$

Therefore the A9 arbitrary-source frozen-shell theorem applies conditionally to its PSD quadratic block, and its Schatten estimate gives

$$\|T_{j, B_{j-1}^{(J)}}(z)\|_{\mathfrak{S}_2}^2 \leq C 2^{-j} (\|z\|_4^4 + L^3 \kappa_0^2). \quad (7.2)$$

This is a valid reusable summable determinant cost.

It does not bound (6.4). Once the present shell has been conditioned out, $w_{A,i,r}$ and $\Delta w_{A,i,r}$ depend on the same past noise and adapted control. A9's base-Gaussian L^2 estimates do not control every such tilted law; direct Malliavin integration by parts also creates a derivative of the control which the Boue–Dupuis cost does not bound. A new global relative-form or backward log-Laplace theorem for the full Cartan secant is therefore load-bearing. Rewriting (6.4) as entropy, HJB, or a Follmer drift without a new estimate is circular.

8 Devil’s-advocate review

1. **“The uniform $C^{1,1}$ heat estimate makes the shell sum summable.” UPHELD as false.** The scalar identity (3.5) saturates the scale-zero contraction; the production computation (3.7) shows the same plateau.
2. **“The coefficient curl means the full shell score is not a gradient.” DISMISSED.** Equation (1.4) is the exact energy gradient. The curl survives only in the internal current-frame decomposition and reappears correctly in (4.8).
3. **“The source factor four disproves the route.” DISMISSED.** Equation (1.19) shows exact neutralization by the nonlinear remainder and retained square. It disproves only the charge-alone and terminal-only readings.
4. **“The Gibbs–Pythagoras formula proves the Nelson bound.” UPHELD as false.** The sequential normalizers in (1.15) are past-dependent. Bounding their sum, or equivalently (6.4), is the open theorem.
5. **“The heat dummy r is the actual future Gaussian field.” UPHELD as false.** In (4.3)–(4.8), r is held spatially constant. Identifying it pathwise with a field would omit its derivative and corrupt the Cartan formula.
6. **“Closing the low block and trace tail closes one-use.” UPHELD as false.** The raw-current secant (6.4) remains at critical regularity. Deleting its positive square or applying (6.6) repeats the forbidden past-energy payment.
7. **“The ordered-shell proof automatically covers every progressive control.” UPHELD as false.** A flow that revisits a Fourier range can cancel an earlier field in that range and requires a separate reduction. The present theorem states the orthogonal one-shot block hypothesis explicitly.
8. **“The numerical plateau proves continuum nonexistence.” UPHELD as false.** It is a regression test and architecture no-go. No nonexistence theorem, regulator removal, or physical phase claim follows.

9 Reproduction and references

The fail-closed command is

```
python codes/foundations/  
a13_classii_backward_heat_martingale_square_coupled_cartan_reduction_verify.py
```

The primary route checks the exact martingale/telescope, square and nonlinear partition identities, full production frame secant, scalar no-gos, and the $N = 64, 128$ production plateau with order-seven/order-nine convergence. The non-importing route defines the Pauli frames independently, uses different quadrature and seeds, verifies the triangular state-past pullback, and computes the isotropic production coefficient. Both write JSON artifacts and the integrated verifier pins all source, note, PDF, claim, gate, and ledger boundaries.

The stochastic variational framework is due to M. Boue and P. Dupuis, “A variational representation for certain functionals of Brownian motion,” *Annals of Probability* 26 (1998), <https://doi.org/10.1214/aop/1022855876>. A related scale-flow use in constructive field theory is N. Barashkov and M. Gubinelli, “A variational method for Φ_3^4 ,” <https://arxiv.org/abs/1805.10814>. These references supply method, not the model-specific identities or the remaining estimate (6.4).

Result footer

Result ID: A13-CLASSII-BACKWARD-HEAT-MARTINGALE-SQUARE-COUPLED-CARTAN-REDUCTION
Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION
Tier: T4 before / T4 after (no promotion)
Closed: full score and retained-square charge; Gibbs-Pythagoras;
terminal-backward heat martingale; controlled telescope;
shellwise heat/charge no-gos; averaged frame/Cartan secant;
finite low block; summable high trace-secant loss
Open: A13-CLASSII-AVERAGED-RAW-CURRENT-CARTAN-JACOBI-FORM-BOUND
Umbrella: A13-CLASSII-CONTROLLED-SHELL-ENERGY-ONE-USE remains open
Scope: finite cutoff; fixed floor; common real-even mutually orthogonal
one-shot sharp shells; regular strict-past block controls
No claim: arbitrary finite-energy drift extension; Nelson estimate;
interacting measure; removal limits; phase transition; BCC;
T5, T6, or T7

Final boundary. The apparent heat-current divergence has been cancelled exactly, and all covariance-trace channels have been reduced below arbitrary one-use budgets. Within the regular control class, the sole load-bearing analytic remainder is the global averaged raw-current Cartan–Jacobi form (6.4), with its positive square retained. Extension to arbitrary finite-energy drifts also remains open.